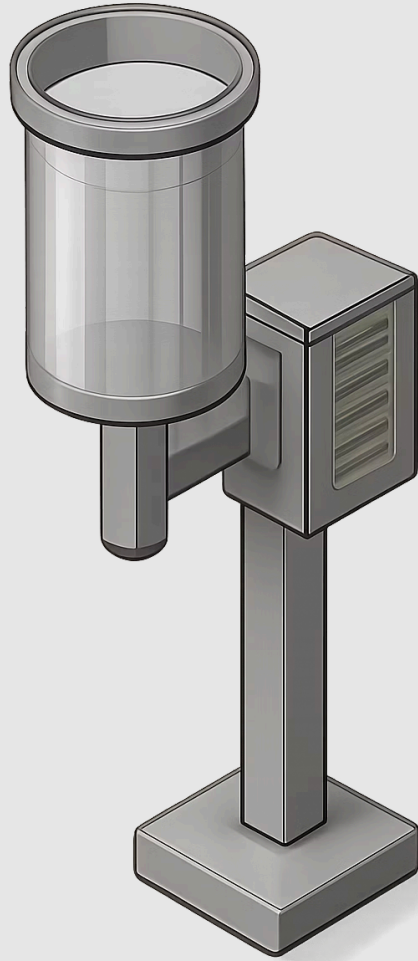




PRAICP-1004: Rainfall Time Series Forecasting

Artificial Intelligence capstone: monthly rainfall at Changi Climate Station (1982-2020). 462 monthly observations. Objective: produce reliable forecasts for the next 12 months — both total monthly rainfall and the highest daily rainfall per month.

Problem Statement & Objectives

Primary objectives

- 1) Forecast total monthly rainfall for next 12 months.
- 2) Forecast highest single-day rainfall within each month for the same horizon.

Supporting goals

Conduct EDA, preprocess time-series data, engineer features, compare statistical and ML models, tune the best model, and extract business insights.

Audience use-cases

Reservoir management, flood preparedness, infrastructure planning and seasonal resource allocation.



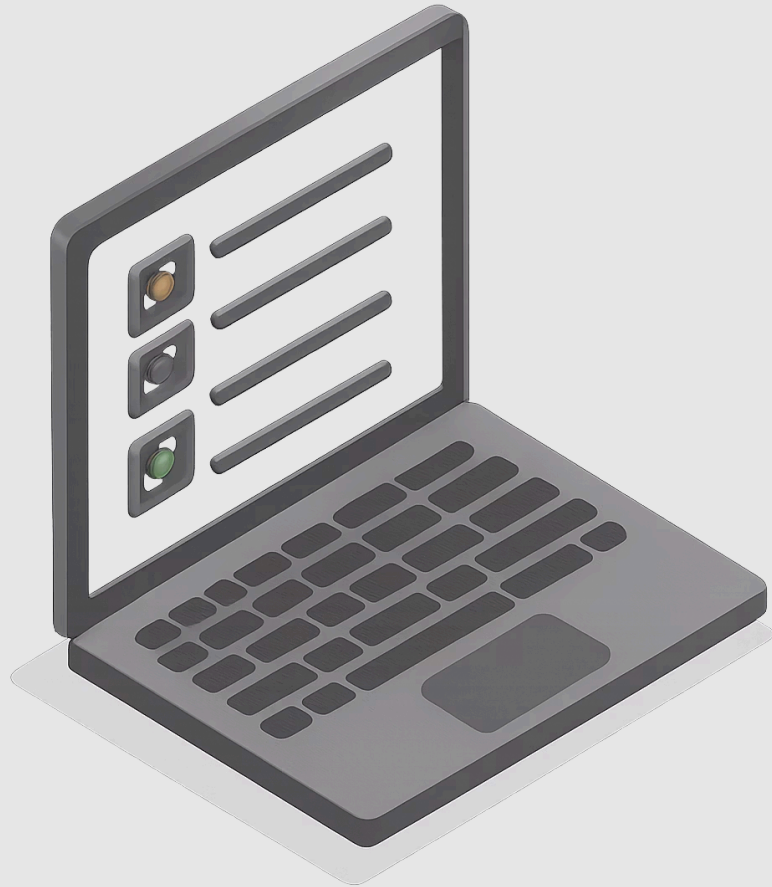
Data & Ingestion



Files merged

- `rainfall-monthly-total.csv` — total monthly rainfall
- `rainfall-monthly-highest-daily-total.csv` — monthly max daily rainfall
- `rainfall-monthly-number-of-rain-days.csv` — rainy days per month

Process: extract ZIP → read CSVs → merge on month → convert month → datetime → sort → set monthly DateTime index (freq = M).



Data Quality Checks (cell-wise)

01

Initial inspection

`head()`, `tail()`, `shape`, `columns`,
`dtypes`, `info()` and `describe()` to
validate format and value ranges.

02

Integrity checks

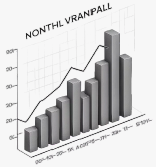
Missing values, duplicates,
start/end dates and expected
monthly calendar comparison (no
missing months detected).

03

Decision

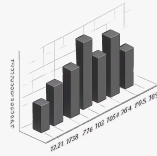
No imputation required — chronological continuity verified, proceed to
EDA and feature engineering.

Exploratory Data Analysis — Key Visuals



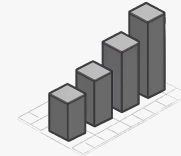
Historical trend

Monthly total rainfall vs time — reveals long-term trend, seasonality and extreme events (spikes).



Seasonality

Average by calendar month — confirms a 12-month repeating seasonal pattern (monsoon cycles).



Distributions

Histograms for total, max-daily and rainy-day counts show skew and frequency of extreme values.

Decomposition & Stationarity

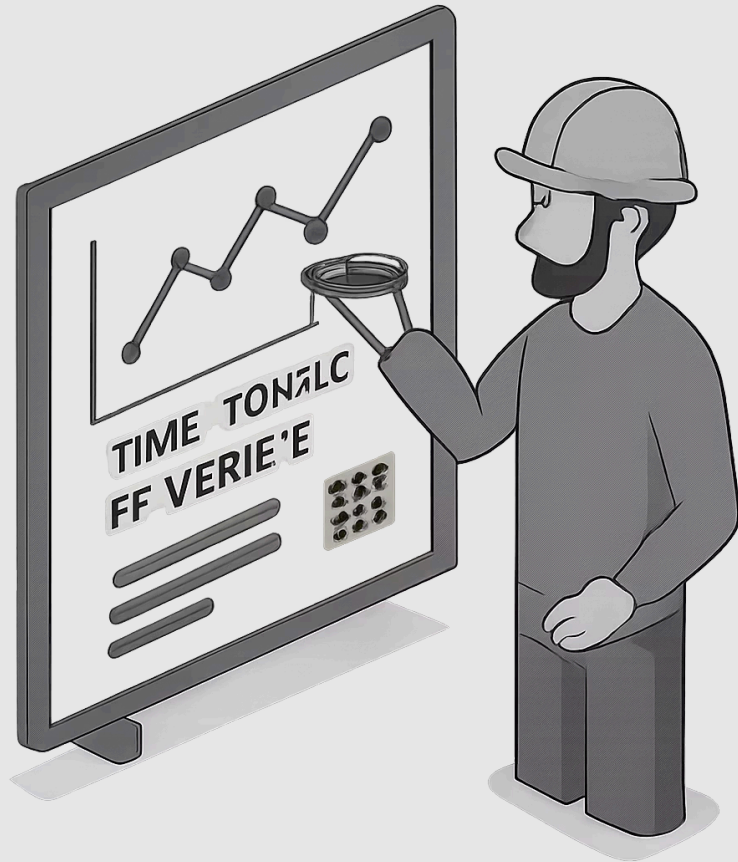


The decomposition isolates long-term trend, repeating 12-month seasonality and residual noise for model selection.

Stationarity check

Augmented Dickey-Fuller (ADF) test used to assess stationarity. If non-stationary, apply differencing prior to SARIMA or use trend-removal for ML features.

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Feature Engineering for ML

1

Time features

month_num, year — encode seasonal position and long-term progression.

2

Lag features

lag_1, lag_2, lag_3, lag_12 — capture short-term autocorrelation and yearly seasonality.

3

Rolling statistics

rolling_mean_12 — smooths past year behaviour to provide stable seasonal context.

Models & Evaluation Strategy



SARIMA

Seasonal statistical model (captures AR, I, MA + seasonality). Strong candidate given 12-month cycle.



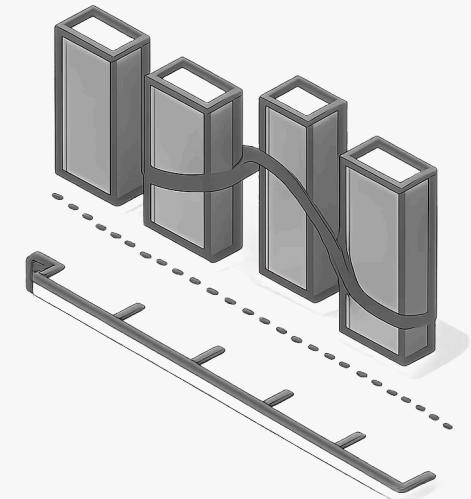
ML regressors

Linear Regression, Random Forest, Gradient Boosting, XGBoost — use engineered features to predict next-step values.

Evaluation & Tuning

Chronological train/test split (no shuffle). Metrics: MAE, RMSE, R^2 .

Hyperparameter tuning with GridSearchCV using TimeSeriesSplit to avoid leakage.



Multi-step Forecasting Approaches



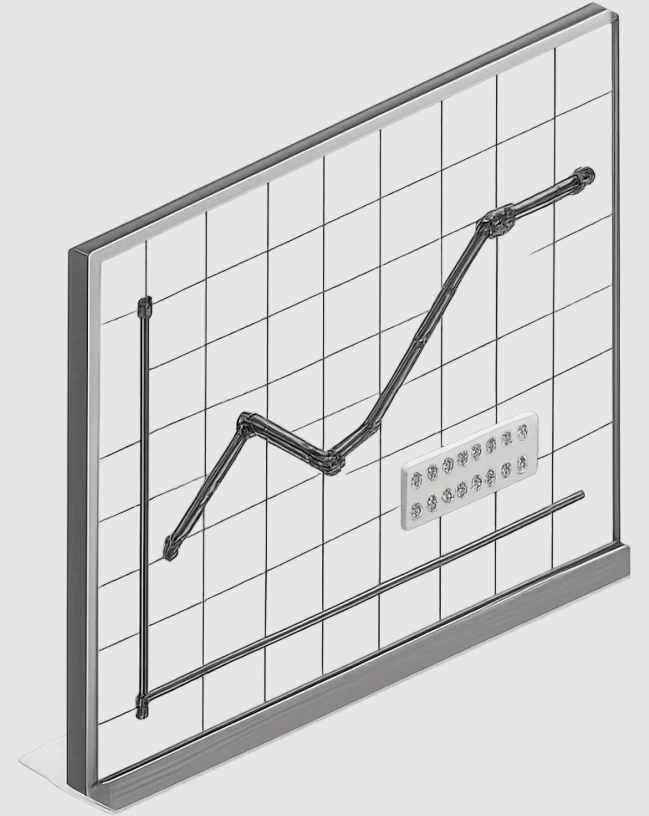
SARIMA (direct multi-step)

Refit on full history and forecast 12 future months directly from the model's seasonal structure.



ML: Recursive forecasting

One-step model used recursively: predict month $t+1 \rightarrow$ feed prediction as lag to produce $t+2 \dots$ until month $t+12$. Be aware of error accumulation.



Results, Business Insights & Next Steps

Key findings

SARIMA captured seasonal cycles most reliably; ML models offered flexible nonlinear fits but required careful recursive forecasting and tuning.

Business impact

Use forecasts for reservoir management, flood preparedness, infrastructure scheduling and seasonal planning. Highest-daily forecasts support flash-flood risk mitigation.

Future enhancements

Move to daily data, add external climate indices (ENSO), automate pipeline and retrain models, deploy forecasting API for operational use.

📄 Viva-ready summary: Load → Preprocess → EDA → Stationarity & decomposition → Feature engineering → SARIMA + 4 ML models → TimeSeriesSplit tuning → 12-month forecasts (totals + highest daily)